

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Examination – 2023

Course: B. Tech. Branch : Electronics & Computer Engineering Semester :IV

Subject Code & Name: Database Management System (BTECP402)

Max Marks: 60

Date: 15.07.2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level /
CO) Marks

Q. 1 Solve Any Two of the following.

12

- | | | | |
|-----------|--|---------------|----------|
| A) | With help of a diagram, explain the difference between 2-tier and 3-tier architecture. Justify which architecture is better suited for web applications? | 02/CO1 | 6 |
| B) | Describe with a diagram the level of abstractions available in database systems. | 02/CO1 | 6 |
| C) | Describe the components of a a) Query Processor b) Storage Processor | 02/CO1 | 6 |

Q.2 Solve Any Two of the following.

12

- | | | | |
|-----------|---|---------------|----------|
| A) | Write SQL commands,
i. To Create table for following schema: Student {Roll-No (PK), Name, Class, DOB}
ii. To add new column percentage to student table.
iii. To drop column DOB from student table. | 03/CO2 | 6 |
| B) | Consider an EMPLOYEE table having Eid, Name, Age, Address, and Salary.
Using subqueries, write a query to
i) Display all details where Salary>45000
ii) Delete all the records of employee whose age<20.
iii) Display name & salary of employee having second highest salary. | 03/CO2 | 6 |
| C) | Explain Inner, Left Outer and Right Outer Joins with suitable example. | 02/CO2 | 6 |

Q. 3 Solve Any Two of the following. **12**

- A) What is Normalization? Explain various anomalies with suitable example. **02/CO3 6**
- B) A company has following entities and respective attributes, where EmpPhone is a multivalued and Address is a composite attribute. Draw the ER-Diagram for the same;

Entity	Attributes	03/CO3	6
Employee	EmpId, EmpName, Gender, DoJ, Designation, EmpSal		
Contact Details	EmpPhone, EmpMailID, Address		
Department	DepId, DepName, DepLocation, DepHead.		
Project	ProjId, ProjName, ProjLocation, ProjDuration		

- C) Solve the following
- a. Find the Primary & Non-Primary Attributes of **03/CO3 6**
R (A,B,C,D,E) FD: {A → BCDE, BC → ACE, D → E}
- b. Identity whether the given equations is in 2NF or not
R (A,B,C,D) FD: {AB → CD, C → A, D → B}

Q.4 Solve Any Two of the following. **12**

- A) Draw a state diagram and discuss the typical states that a transaction goes through during execution. **03/CO4 6**
- B) What is concurrency control? Explain lock based protocols for concurrency control. **02/CO4 6**
- C) Describe shortly **02/CO4 6**
a) Database Recovery Techniques
b) Causes of Database failures

Q. 5 Solve Any Two of the following. **12**

- A) State the CAP theorem. Explain each term of CAP with suitable example. **01/CO5 6**
- B) Describe the concept of Document and Collection in MongoDB. **02/CO5 6**
- C) For MongoDB, write a query to **03/CO5 6**
a) Create a collection named **Library**
b) Create a document named using data given below
BN101, "Database System Concepts", "Henry Korth", 800.50.
c) Insert the following book details using single query.
BN102, "Core Java", "Nageswara Rao", 650.75.
BN103, "Let Us C", "Balaguruswamy", 400.50.
BN104, "DBMS", "G.D. Patil", 250.5
d) Retrieve / Display all the above documents
e) Update the Author name to "Y Kanetkar" for BookId BN103
f) Delete a document with BookId BN104.

***** End *****